

Seukwoo Lee 이석우

AI Engineering Lead · Development leader, 10 years

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In one line

A development leader with 10 years of experience, turning research- and pilot-stage technology into products people actually use and pay for. I design and write the code myself, and have led teams from a 3-person dev unit to a 50-person research division.

Core strengths

01 From pilot technology to commercial product

- Led Alan's monetization as PO, planning to launch — Pro launch (2025.07), first paid revenue (not disclosed)
- Built Dr.Meta and Meta.CRO on MX Studio output — used at cancer centers nationwide

02 Designing AI systems hands-on, proving them in code

- Designed a rules + AI code-generation pipeline and built its core (PoC) → launched with the team as ProtoPie MCP (Oct 2026)
- Built Decision Graph solo — extracts decisions and auto-graphs their relations; 80%+ of human-labeled decisions found

03 Deciding by measurement, not gut feel

- Chose models by testing (lighter model rejected at 17/24) and cut calls 74%, cost 25%
- Set development priorities from traffic flows and per-feature usage in Google Analytics and Amplitude

04 Leadership sized to the team

- Led teams from 3 people to a 50-person R&D division — acting technical lead for a year without a CTO; 3 group awards
- Running an agile team of 4–5 AI engineers and app developers — build small, validate fast
- Evaluated members and leads, and adjusted how I lead from feedback both ways

Experience

Studio XID (ProtoPie) · Head of Product Dev Team

2026.01 - Present

Product Dev Team

Mission Take design-to-code AI from PoC to official launch — completed with the ProtoPie MCP launch (2026.10)

LLM pipeline for decision extraction (Slack·Notion → decision records with evidence) 2026.08 - 2026.10

Design & development (solo) · New product PoC: Decision Graph

- Hands-on Turned PO requirements into a tech spec; built solo an LLM PoC extracting who decided what and why, with evidence
- Hands-on Auto-visualized decisions as a relation map and timeline, each linked to its source text
- Hands-on Validated on a human-labeled golden set from one real project — 80%+ of labeled decisions extracted
- Hands-on Tested key risks in the PoC: human ground truth, one-way records, no duplicates on re-runs
- Hands-on Two-tier inference (light model triages, strong model extracts/verifies), chosen by tests; all-light 17/24

- Hands-on Profiled per-request cost: 74% fewer calls (1,289 → 336), 25% lower cost ($10.2 → $7.6); incremental $0.7
- Hands-on Checked every change against 24 test cases, with a cap on evaluation cost

New HarmonyOS app launch (ported from the Android app, Huawei AppGallery) 2026.06 - 2026.08

Product Owner · Product: ProtoPie Player (HarmonyOS)

- Lead As PO with one developer, shipped in 2 months — engine, renderer, UI rewritten via spec-driven dev and AI agents

AI-based code generation system (PoC → productization) 2026.01 - 2026.10

Head of Product Dev Team · Product: ProtoPie MCP (Dev View · Code MCP) official launch

- Hands-on Designed the rules + AI-stage code generation pipeline and built its core (PoC)
- Hands-on Designed training for image/layout-understanding models and built the training data
- Hands-on Designed how the pipeline integrates into the existing product (ProtoPie)
- Lead Ran a 4–5 person agile team — AI engineers trained the models; integration built with the team
- Lead Designed the MLOps pipeline from data cleaning → training → deployment
- Lead Expanded to AOS, iOS and Web; launched as ProtoPie MCP in Oct 2026 (7 frameworks)

ESTsoft · Product Owner 2025.04 - 2026.01

AI Agent Lab / Alan Development Team / PO

Mission Turn the pilot service Alan into a paid product — done with the Pro launch (2025.07) and first paid revenue

AI search engine service (LLM-based agentic AI) Advancing 'Alan' (ESTsoft) 2025.04 - 2026.01

Product Owner · Product: Alan

- Lead As PO, took a research pilot service to commercial launch and monetization
- Lead Ran the AI researcher and FE/BE developer team — feature planning, schedule and risk
- Lead Architected on Azure, FastAPI, React and LangGraph; separate Dev/QA/Stage/Release environments
- Lead Built in-house agents for image search, YouTube summaries and report generation
- Hands-on Planned slide generation — Genspark benchmark, user scenarios and generation flow
- Hands-on Designed the staged slide agent (intent→scope→research→style→HTML→revision), human-in-the-loop
- Hands-on Planned AI monetization: priced Pro and usage limits from cost per call, competitors and conversion
- Lead Paid launch — entitlements, payment/cancel/refund, terms and notices (2025.07, announcement)
- Collab QA, auth, payments and infra teams for testing; marketing and business for the launch
- Lead Shipped deep research, slide generation and a back-office tool; live at myalan.ai
- First paid revenue from Pro (not disclosed); then planned an enterprise product → institutional contracts

Tmax Group · 8 yrs 3 mos 2017.01 - 2025.04

TmaxMetaAI · Project Leader, Development Team 2022.07 - 2025.04

Meta AI Research Division / Researcher (Head of Office)

> **Software for building 3D web components** 'MX studio' and productization on it 2022.07 - 2025.02
>
> Project Leader · Product: MX studio
>
> - Hands-on Engine/renderer design — physics, real-time rendering, 3D events/actions, node-based visual code
> - Lead Engine team and task force; productization (frontend UI/UX, backend, auth)

- Lead Productized Dr.Meta and Meta.CRO on its output; external beta

Metaverse platform used by cancer centers nationwide Development of 'Dr.Meta' (Korea Smart Healthcare Association) 2024.06 - 2025.01

Tech Lead · Product: Dr.Meta

- Lead Tech Lead (separate PM) — React web app ↔ Unity app integration, admin/permission features
- In commercial operation at cancer centers nationwide (web.drmeta.kr)

3D-based virtual simulation training platform for clinical trials Development of 'Meta.CRO' (Korea Smart Healthcare Association) 2023.06 - 2025.01

Tech Lead · Product: Meta.CRO

- Lead Tech review, design, staffing and dev management (separate PM); 3D render library (private npm)
- In commercial operation at cancer centers nationwide

Online marketplace platform for contact lenses Development of 'NaenunN' (Pixelro) 2024.03 - 2024.09

Project Leader · Product: NaenunN

- First online contact-lens sales service (regulatory sandbox) — shipped WebView iOS/Android apps, led payment/delivery API and admin design; in operation

GIS-based Samsung data center information system Development of 'GIS S-DCIS' (Samsung C&T) 2023.12 - 2024.08

Project Leader · Product: GIS S-DCIS

- Lead GIS system on GeoServer · PostGIS · OpenLayers for a closed on-premise network — co-developed and delivered with Yeonwoo Technology (B2B)

Real-time multi-sync server Research on 'MVS: Metaverse Server' (internal project) 2023.01 - 2024.09

Tech Lead · Product: MVS

- Lead Tech review, design and dev management of a C++ real-time sync server — used for Dr.Meta

TmaxAI · Project Leader 2020.07 - 2022.06

Meta Vision Research Division / Researcher (Head of Office, concurrently Division Head)

- R&D Division Head (concurrently Head of Office) — led an R&D organization of up to 50
- Acted as technical lead for about a year while the CTO role was vacant
- Ran evaluations and feedback for team members and team leads

Metaverse education platform Development of 'Mint' 2021.07 - 2022.06

Tech Lead · Project: Mint

- Lead Tech review, design and dev management of a metaverse education platform — Kongju National Univ.

Advanced technology research Deep Learning, 3DMM, SLAM 2020.07 - 2021.06

Researcher · Research: Digital human and metaverse space research

- Collab With teammates, researched deep-learning digital humans (TTS) and SLAM 3D spaces — demoed at Tmax Day 2020

TmaxA&C · Product Engineer 2017.01 - 2020.07

Graphics Research Division / Researcher (Team Lead)

Collaboration solution Development of 'WAPL' 2019.01 - 2020.06

Product Engineer · Product: WAPL

- Frontend development of the collaboration solution's calendar and messenger — commercialized by TmaxWAPL

Linux-based product Development of 'TmaxOS' 2017.06 - 2018.12

Product Engineer · Product: TmaxOS

- C++ development for a Linux-based OS — built and refactored the settings screen and video player

Integrated system interface solution software Structure analysis of 'Anylink' 2017.02 - 2017.06

Structure analysis & refactoring design · Product: Anylink

- Collab Structure analysis and refactoring design of an integration solution — a TmaxSoft commercial product

Skills

AI · LLM — Python · LLM API · LangGraph · MCP · Computer Vision

Backend · Infra — Node.js · FastAPI · Java · Spring · C/C++ · Docker · Jenkins · AWS · Azure

Frontend · Cross-platform — TypeScript · JavaScript · React · React Native · Three.js · Unity

Collaboration · Analytics — Git · Jira · Notion · Figma · Google Analytics · Amplitude

Platforms — Windows · Linux · iOS · Android

Education

KAIST, Culture Technology · M.S. in Culture Technology (Master's degree) 2013.09 - 2016.02 (Graduated)

3.5 / 4.3 · Human-computer interaction, computational design, game studies, IoT

Ajou University, Digital Media · B.S. in Digital Media (Bachelor's degree) 2007.03 - 2013.08 (Graduated)

3.76 / 4.5 · Computer programming, computer architecture, object-oriented programming, information design

Awards & activities

Award

2023.01 **Annual Technology Innovation Award** — group-wide, once a year, about 10 people

2022.06 **Super Leader Award** — group-wide, twice a year, about 5 people

2022.01 **Maestro Award** — group-wide, once a year, about 3 people

Paper & research

2016.02 **Master's thesis** — A Study on Mobile Game Design Strategies that Enhance Enjoyment and Self-Efficacy to Promote Everyday Physical Activity

2016.01.29 **Conference paper 2** — A Study on Mobile Game Design Using Gamification and Symbolization Strategies to Promote Physical Activity in Everyday Life

2016.01.28 **Conference paper 1** — Gamification Can Affect the Sustainability of The Wearable Device for Physical Activity? : User Case Study

2014.11 **Journal paper** — An Exploratory Study of Factors Affecting the Usability and Sustainability of Wearable Devices for Health Management - A Review of User Experience from a Convergence Perspective of Technology, Psychology and Interaction

Certifications & languages

2025.01.12 **OPic Speaking** — English (IM1) · ACTFL

2010.09.03 **MOS Master 2007** — Microsoft Corporation